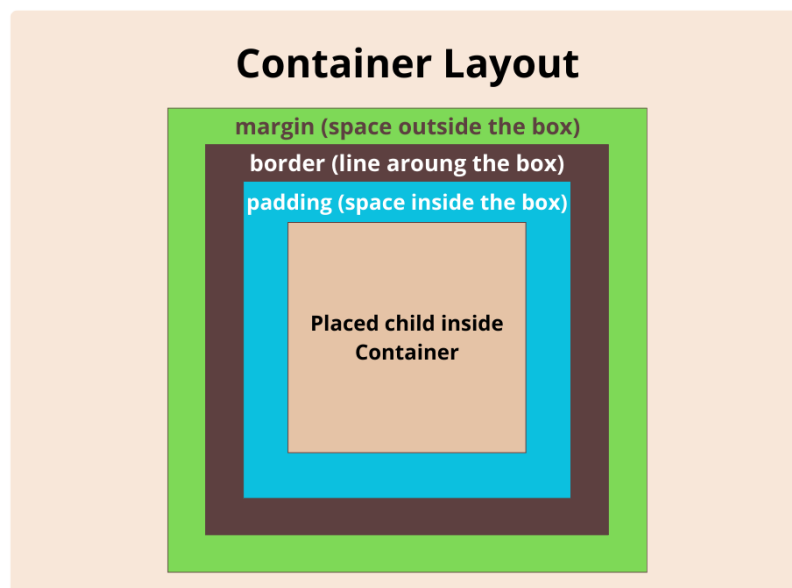


- Flexible for any layout or UI design.

Container Widget

- What is the Container Widget?
 - A customizable “box” used to style, position, and layout elements in an app.
 - Often used to wrap other widgets and apply spacing, colors, or decorations.
 - One of the most flexible and commonly used widgets in Flutter.
- Key Features of Container
 - Commonly used properties:

Property	Description	Example
margin	Adds space outside the container.	margin: EdgeInsets.all(10)
padding	Adds space inside the container.	padding: EdgeInsets.all(15)
width & height	Sets the size of the container.	width: 200, height: 100
color	Sets the background color.	color: Colors.blue
decoration	Adds styling like borders, gradients, and shadows.	decoration: BoxDecoration(...)



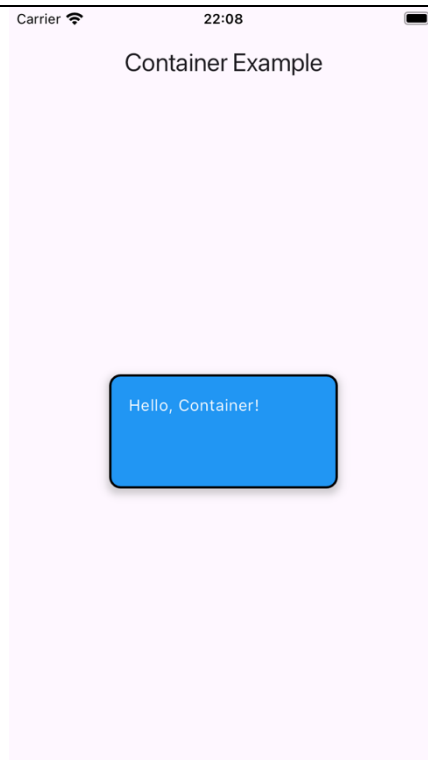
- Customizing Container with BoxDecoration
 - Additional styling options with BoxDecoration:
 - color → Sets the background color.
 - border → Adds a border around the container.
 - borderRadius → Rounds the corners of the container.
 - boxShadow → Adds a shadow effect.
 - gradient → Creates a gradient background.
- Example: Styled Container in Flutter
 - Code Example:

```
Container(
  width: 200, // Width of the container
```

```

    height: 100,          // Height of the container
    margin: EdgeInsets.all(10), // Space outside the
container
    padding: EdgeInsets.all(15), // Space inside the
container
    decoration: BoxDecoration(
      color: Colors.blue,          // Background color
      border: Border.all(          // Adds a border
        color: Colors.black,
        width: 2,
      ),
      borderRadius: BorderRadius.circular(10), // Rounded
corners
      boxShadow: [                // Adds shadow
        BoxShadow(
          color: Colors.grey.withOpacity(0.5),
          spreadRadius: 2,
          blurRadius: 5,
          offset: Offset(0, 3), // Shadow position
        ),
      ],
    ),
    child: Text(
      'Hello, Container!', // Child widget inside the
container
      style: TextStyle(color: Colors.white),
    ),
  ),
)

```



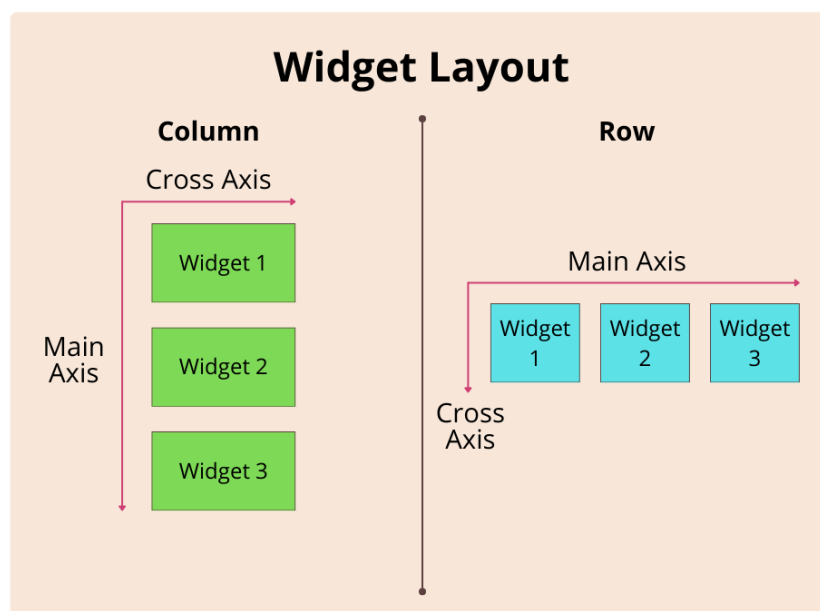
- How It Works:

- width & height → Sets container size.
 - margin: EdgeInsets.all(10) → Adds space outside.
 - padding: EdgeInsets.all(15) → Adds space inside.
 - decoration → Applies background, border, shadow, and rounded corners.
 - child → Contains a Text widget inside the container.
- e. Why Use the Container Widget?
- Main Benefits:
 - Adds spacing with margin and padding.
 - Styles elements with borders, colors, and shadows.
 - Wraps content to control size and positioning.

Row and Column Widget

- a. What are Row & Column?
- Used to arrange widgets in a horizontal or vertical layout.
 - Row → Aligns widgets horizontally (side by side).
 - Column → Aligns widgets vertically (one on top of another).
- b. Understanding Main Axis & Cross Axis
- When using Row or Column, there are two important axes:

Axis Type	Row	Column
Main Axis	Horizontal (左⇌右) → Widgets are placed side by side .	Vertical (上⇕下) → Widgets are placed one below another .
Cross Axis	Vertical (上⇕下) → Controls alignment top, center, bottom .	Horizontal (左⇌右) → Controls alignment left, center, right .



- c. Example: Row & Column in Action